

Let Ω be a bounded piecewise smooth domain in $\mathbb{R}^2$ with oriented boundary curve Γ . Also, let X be an open neighborhood of $\overline{\Omega}$ in $\mathbb{R}^2$, and let $g, h \in C^1(X)$. Then

$$\int_{\Gamma} g dx + h dy = \int_{\Omega} \frac{\partial h}{\partial x} - \frac{\partial g}{\partial y} dx dy.$$

Proof: We prove the generalized Stoke's theorem for manifolds:

Theorem: Let M be an oriented smooth n —manifold with boundary, and ω be a compactly supported smooth $(n - 1)$ —form on M . Then

$$\int_M d\omega = \int_{\partial M} \omega.$$

(here, ∂M has the induced orientation, and ω is $\iota_{\partial M}^* \omega$.)

Proof: We begin with a special case: suppose $M = \mathbb{H}^n$ is the upper half-space, then take $R > 0$ such that $\text{supp} \omega \subset A = [-R, R] \times \cdots \times [-R, R] \times [0, R]$. Write $\omega = \sum_{i=1}^n \omega_i dx^1 \wedge \cdots \wedge \hat{dx}^i \wedge \cdots \wedge dx^n$, then

$$d\omega = \sum_{i=1}^n (-1)^{i-1} \frac{\partial \omega_i}{\partial x^i} dx^1 \wedge \cdots \wedge dx^n.$$

Thus by the fundamental theorem of calculus,

$$\begin{aligned} \int_{\mathbb{H}^n} d\omega &= \sum_{i=1}^n (-1)^{i-1} \int_A \frac{\partial \omega_i}{\partial x^i} dx^1 \wedge \cdots \wedge dx^n \\ &= (-1)^{n-1} \int_{[-R, R]^{n-1}} \omega_n(x^1, \dots, x^{n-1}, 0) dx^1 \cdots dx^{n-1}. \end{aligned}$$

Since the pullback of dx^n to $\partial \mathbb{H}^n$ is zero, the other side becomes

$$\begin{aligned} \int_{\partial \mathbb{H}^n} \omega &= \sum_i \int_{A \cap \partial \mathbb{H}^n} \omega_i(x^1, \dots, x^{n-1}, 0) dx^1 \wedge \cdots \wedge \hat{dx}^i \cdots \wedge dx^n \\ &= \int_{A \cap \partial \mathbb{H}^n} \omega_n(x^1, \dots, x^{n-1}, 0) dx^1 \wedge \cdots \wedge dx^{n-1}. \end{aligned}$$

Since $(x^1, \dots, x^{n-1})$ has orientation $(-1)^n$ for $\partial \mathbb{H}^n$, the two sides are equal.

Another special case $M = \mathbb{R}^n$ is analogous, since the right side is zero, and each term of the left side vanishes.

Now let M be an arbitrary smooth manifold (with boundary), and consider a special $(n - 1)$ —form ω such that $\text{supp} \omega \subset (U, \varphi)$ for some local chart. Assuming that φ is positively orientated boundary chart, by definition

$$\int_M d\omega = \int_{\mathbb{H}^n} (\varphi^{-1})^* d\omega = \int_{\mathbb{H}^n} d((\varphi^{-1})^* \omega)$$

and by the computation above, this becomes

$$\int_{\partial \mathbb{H}^n} (\varphi^{-1})^* \omega.$$

Clearly $\varphi|_{U \cap \partial M}$ is an orientation-preserving diffeomorphism onto $\varphi(U) \cap \partial \mathbb{H}^n$ hence it is equal to $\int_{\partial M} \omega$. It is analogous for negatively oriented φ , and for interior charts with $\mathbb{H}^n$ replaced by $\mathbb{R}^n$.

Finally, consider an arbitrary compactly supported $(n-1)$ -form ω . Choose a cover of $\text{supp} \omega$ by finitely many domains of charts $\{U_i\}$ and a smooth partition of unity $\{\psi_i\}$. Apply the preceding argument to $\psi_i \omega$ we obtain

$$\int_{\partial M} \omega = \sum_i \int_{\partial M} \psi_i \omega = \sum_i \int_M d(\psi_i \omega) = \sum_i \int_M d\psi_i \wedge \omega + \psi_i d\omega = 0 + \int_M d\omega.$$